

10. All equipment necessary to carry out factory and on site testing and commissioning;
11. All software and hardware required for data logging.
12. The ECS contractor will be responsible for sealing with fire rated material of all openings left due to ECS works.
13. Provision of hook in the roof to facilitate lifting of the equipments for installation. These hooks are to be left for future use by the maintenance branch.
14. All electrical and ventilation work of mid shaft provisions as defined in the BOQ of Mid shaft.

The emphasis is to explain the requirements of work, interfaces with other contractors for achieving an efficient & safe working system commensurate to the best international standards and practices. Every effort has been made to cite the design requirements very clearly, however the contractor shall follow acceptable standards & design procedures akin to the best available in world Metros where this is not explicitly mentioned to ensure a perfect and complete execution of ECS works

1.4 Relevant Documents

This Specification should be read in conjunction with the General Conditions of Contract (GCC), the Special Conditions of Contract (SCC), the Employer's Drawings and any other document forming part of the Contract.

In the event of a conflict between this Specification and any other standards or specification quoted herein, the requirements of this Specification shall prevail.

The order of precedence, with Item (a) having the highest priority, is:

- (a) BOQ
- (b) Specification
- (c) Drawings
- (d) International or Indian Standards referenced herein.

Notwithstanding the precedence specified, the Contractor shall always immediately seek advice from the Employer in the event of conflicts between Specifications.

1.5 Verification and validation of design

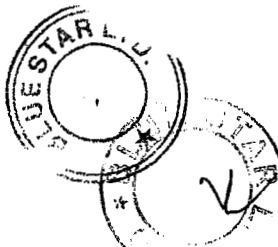
Although responsibility for the design service of the Works lies with the Detailed Design Consultants (DDC) the Contractor shall thoroughly satisfy himself that the tentative capacities, ratings and quantities of equipment as specified herein meet the functional and operational requirements.



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The contract price shall add any necessary equipment, equipment of higher capacities and higher ratings for the systems and sub-systems necessary for the complete, safe, reliable and operable Environment Control System providing all clarifications and justifications for the same.

2 Responsibility of the ECS Contractor

The Contract BE-12 LOT3 & BE-14 LOT3 Contractor shall be responsible for design verification and detailed engineering, Manufacture, Supply, Installation, Testing and Commissioning of ENVIRONMENT CONTROL SYSTEM for for 7 underground stations, tunnel sections and mid shaft provisions in the CS-QM corridor.

2.1 Scope of work

The detailed scope of work for ENVIRONMENT CONTROL SYSTEM (ECS) is described in the specification.

3. Drawings, Documents, Records and Manuals

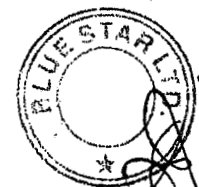
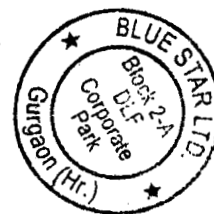
The contractor shall submit in English language, all the following drawings and documents according to the time table defined below which may be modified according to the contractual planning of the equipment supply:

3.1 Within two months after the Notice to Proceed

- one short bill of quantities for the supplies
- overall execution programme
- detailed construction timetable, precisely defining the various equipments at construction stages
- technical proposal for major equipment

3.2 Within four months after the Notice to Proceed

- detailed bill of quantities for the supplies
- updated execution programme
- shop drawings developed for all elements of construction purpose
- method statement for various activities
- manufacturing and delivery programme for major equipment
- testing and commissioning procedure



3.3 Drawings and Records

The Contractor shall provide 4 copies of drawings, 3 in A1 size and one in A3 size, bound into circuit books. The Contractor shall ensure that, at each equipment location, an as-built copy of the Site documentation is provided. This documentation shall include as a minimum:

~~Circuit wiring book;~~

~~Equipment mode tables; and~~

~~Operation and maintenance manuals.~~

3.4 Asset Identification

The Contractor shall submit an asset database for review by the Engineer. The database shall contain the complete asset listing for the ENVIRONMENT CONTROL SYSTEM.

The database shall be designed with a minimum of the following information:

asset details;

failure history;

date installed; and

date(s) tested.

All equipment and software, down to the line replaceable unit, shall have a unique Identification number that is capable of being identified electronically and manually.

4. Interfaces

The design and construction of the Delhi Metro Railway, is a complex multidisciplinary project, requiring close interaction and co-ordination between the various designers and builders. With a view to minimising, if not eliminating problems which could arise during construction, it is imperative that the scope of work of different Agencies is defined as accurately as possible and also the responsibilities and obligations of different designers/builders to reckon and provide for the works by other related Agencies are clearly spelt out. The objective of the 'Interfacing document' is to address the above requirement.

4.1 Interfacing Contractors

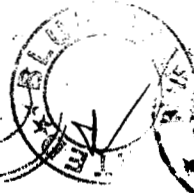
The Contractor will have to interface with the following designated Contractors / Consultants as a minimum during the course of execution as part of Design and Construction interfaces with the contractor for



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- Civil and structural work.
- TVS works under whose scope the supply, installation, testing & commissioning of Trackway exhaust Fan lies.
- Sub-Contractor for architectural finishes,
- Control system Contractor,
- Catenary Contractor.
- Sub Contractor for BMS work for BMS interface.
- Contractor for Electrical and Mechanical works for design of Motor Control Centres and other Electrical Panels.
- Any other contractor as required for execution of complete ECS work.

4.2 Interfacing Agencies

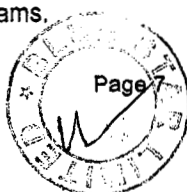
The Contractor may have to interface with the other local and statutory authorities for completing all the works as per Contract. The Contractor shall be responsible for obtaining any "clearances or certificates" from local authorities, if necessary, to make the ECS system fully functional and operational as per the Contract.

4.3 Interface with civil Contractor

Civil construction works of station are to be executed by a separate Civil Contractor. The ECS Contractor shall ensure efficient Interface and Coordination with Civil Contractor concerning ECS works, etc on site..

Such coordination responsibility of the contractor shall include the following:

- (i) To obtain from the Civil Contractor information reasonably required for enabling the Contractor to meet the construction target dates.
- (ii) The Civil Contractor will be the coordinating entity and play the major role in the interface with ECS Contractor, however the ECS Contractor present on site will also be responsible for any delay on the schedule as specified in his contract.
- (iii) ECS Contractor will make sure that he provides the updated valid documents, for the reference of the Civil Contractor in time, where ECS Contractor requires the Civil Contractor to execute his work as per the requirement of the ECS Contractor. These documents will be the reference documents for the Interface Management being carried out by the Civil Contractor.
- (iv) Where the execution of the Civil Contractor depends upon the Site Management or information to be given by the ECS Contractor, the ECS Contractor shall provide correct and accurate information in time so as to enable them to meet their respective programs.



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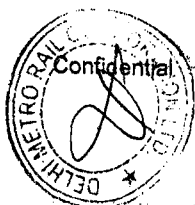
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- (v) It is to be ensured that all provisions for access and delivery of plant is coordinated with and reflected in the ECS Contractors Delivery Route Drawings. The ECS Contractor shall also ensure that all plants are delivered at the time agreed to allow openings to be left in the structure for such delivery in accordance with the Civil Contractor's programme.
- (vi) ECS Contractor must ensure that the production of Working Drawings for Services to be provided in the base slab, other slabs and structures (such as earthing, provision of ducting, conduiting, cable routing, DB mounting, piping and other works Included in the tender) is carried out in time and approval obtained from the Employers Representative. A Copy of Drawings along with schedule for execution of such works must be handed over to Civil Engineering Contractor in time, to enable him to plan his activities. Further, ECS Contractor shall ensure that Information required by Civil Contractor for any concreting / other works where ECS and electrical works are involved is made available in time.
- (vii) The ECS Contractor shall conduct regular meetings with the Civil Contractor as necessary to clarify particular aspects of the interfacing requirements of the works. He will also attend regular Co-ordination meetings convened by the Engineer for interface.
- (viii) The ECS Contractor, shall in carrying out his co-ordination responsibilities, raise in good time and provide sufficient information for the Engineer to decide on any disagreement with Civil Contractor. If the Contractor despite having taken all reasonable efforts cannot resolve such disagreement, then the decision of the Engineer shall be final.
- (ix) The ECS Contractor shall ensure the presence of his qualified and experienced Engineer during Civil construction of station to enable proper interface with Civil Contractor so as to ensure smooth completion of works.
- (x) Access will be provided to the staff of the ECS Contractor for carrying out their works and bringing materials and Equipments at the site. However the security of material and Equipments brought at the site, by the ECS Contractor is ECS Contractor's responsibility.

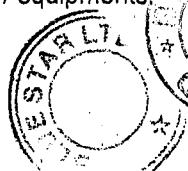
5. Installation

The following paras provide general guidelines for the installation of ECS systems under this Contract. The detailed Installation requirements have been described in the technical specifications for Individual system / equipments



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5.1 Installation Plan and Programme

The Contractor shall prepare and submit the Installation Plan for review of the Engineer 30 days prior to the start of Installation on Site. The Installation Plan shall show how the Contractor proposes to organise and carry out the Installation and complete the whole of the Works by the given Key Dates. The Contractor shall attend weekly planning meetings with the Engineer to finalise the work detail, commencing 4 weeks prior to the start of Installation on Site. **At every site contractor shall display the plan & programme.** The programme shall be submitted in soft & hard copy using primavera or MSP software.

5.2 Method Statement

The Method Statement shall be submitted to the Engineer's for review at least 15 days prior to the installation activity commencing On-Site. This shall show in particular the loadings and modes of transport of the items of major equipment and the routing used as they are taken to their final locations.

Prior to proceeding with Installation, the Contractor shall submit for the Engineer's consent four copies of detailed (shop) drawings showing all installations including dimensions, supports, hardware, installation methods, and all other pertinent data.

The manufacturer's rigging or erection instructions shall be carefully followed. The Contractor shall make certain that the installation of all supports, gadgets, hardware, etc., are accomplished with precision and ensure exercise of extreme care so as to assure safe, accurate and trouble-free installation. Installation shall be undertaken in the presence of the Manufacturer's Field Service Representative or authorised Supervisor. Equipment that is improperly installed shall be removed, checked / tested and reinstalled. Any damage caused due to improper installation and removal shall be rectified before reinstalling at no extra cost.

5.3 Contractor's Resident Staff

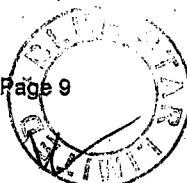
The Contractor shall ensure that a qualified Representative of the manufacturer is available on-Site for the duration of the On-Site Works during normal working hours and installation period and on-call to arrive on Site within 30 minutes at all other times. A manufacturer's representative shall support the Contractor's Representative during the Installation and Testing phase of the Works.

The Contractor's Representative shall have sufficient authority to progress the Contractor's work on Site. The Contractor's Representative shall be competent and qualified to act on behalf of the Contractor, and provide upon request information that may be required.

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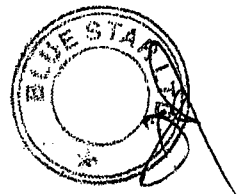
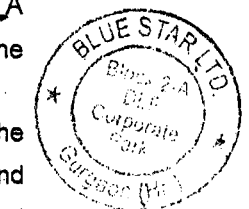


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- Current progress of the Works;
- Planned work for the next 3 weeks;
- Documents and records pertaining to the Works.

6. Testing and Commissioning

6.1 General

The Contractor shall perform all forms of test procedures applicable to the system and shall conduct factory, site installation and acceptance tests.

The commissioning activity shall include a period of Integrated System Testing followed by a period of Trial Running for staff training and familiarisation and timetable proving purposes.

6.2 Test Programmes and Procedures

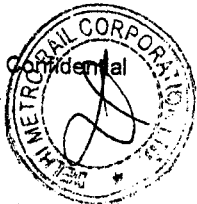
Unless agreed in writing by the Engineer, personnel engaged on testing shall be independent of those directly engaged in the design or installation of the same equipment. All Test equipment shall carry an appropriate and valid calibration label. They shall be periodically checked for calibration accuracy

The Contractor shall sign all reports of Tests. The Contractor shall present a comprehensive Testing and Commissioning Programmes to ensure a completely safe and operable system within 12 months from the Date of Commencement of works.

All Test procedures shall be submitted at least 30 days prior to conducting any Test. Test procedures shall show the extent of testing covered by each submission, the method of testing, Acceptance Criteria, the relevant drawing (or modification) status, and the location. Test Procedures shall be amended, as required, by the Contractor during the currency of the contract to reflect changes in system design or the identification of additional testing requirements.

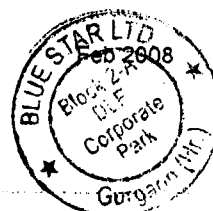
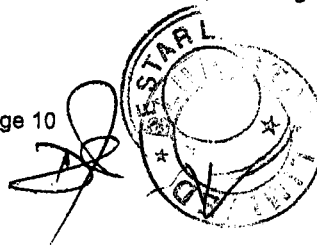
The Employer, the Engineer and/or any of their staff shall have the facility to monitor all Tests and have access to all Test records. All costs associated with Testing shall be borne by the Contractor, unless otherwise specified, including the services of any specialised personnel or independent assessors. The Contractor shall also bear any expenses incurred due to re-testing caused by defects or failure of equipment to meet the requirements of the Contract in the first instance.

The Contractor is reminded that, at some point, the traction system will be energised and that additional precautions for the safety of staff and co-ordination of activities after "power-on" shall be anticipated in his installation, testing and commissioning programmes.



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6.3 Installation Tests

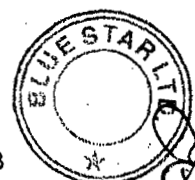
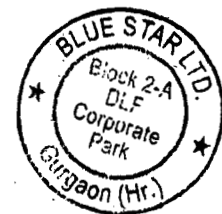
A. Prerequisites for Installation

- Prior to installation, the Contractor shall ensure that equipment delivered to Site has not been damaged in transit and ensure for their dimensional accuracy.
- Designs for the Sections under test shall be completed and submitted to the Engineer for review prior to Installation.

B. Inspection

- The inspection shall verify that equipment has been installed to the procedures and design that have been reviewed without objection by the Engineer and that equipment is correctly located and labelled.
- The Contractor shall maintain inspection records to demonstrate that each item of equipment has been inspected and found to be satisfactory, and attach to this record a detailed list of any discrepancies found and remedial work carried out.

— * — * —



C. Installation Tests

Installation tests shall be carried out by the Contractor for each subsystem following Installation but before Functional Tests to demonstrate that the installation has been carried out correctly and equipment is properly housed and fixed.

During and on completion of installation, the Contractor shall undertake testing of all cables, wiring and equipment, instrumentation and protection devices, in a progressive sequence and in accordance with the overall-testing programmes. These tests shall culminate in Functional Tests to verify the correct operation of all apparatus and, where appropriate, correct response to the respective control commands or monitored function.

D. Functional Tests

The functional tests shall be carried out on installed equipment before System Acceptance Tests (SAT) to demonstrate that the Section of the Works operates correctly in accordance with the Specification.

The functional tests shall sequence through all required operations to prove that the System performs in accordance with the Specification and that the local configuration data (for example, control tables) is correct.

G. System Acceptance Tests (SAT)

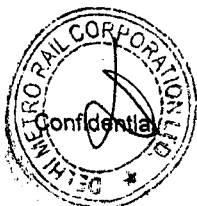
The Contractor shall prepare and organise comprehensive programmes of Tests to demonstrate to the Engineer that all systems, sub-systems and apparatus defined under the Contract meet the specified performance requirements in all respects.

The requirements that shall be satisfied before the commencement of the System Acceptance Tests (SAT) are:

- All documentation pertaining to the Test Report shall be submitted to the Engineer for review.
- Facilities for the maintenance of the System shall be in place.

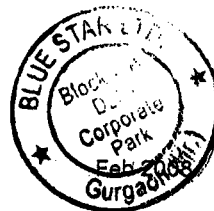
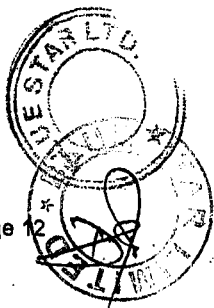
6.4 Integrated Testing

Prior to opening for Public Services, Employer will conduct Integrated System Tests in conjunction with System wide Contractors to demonstrate that the complete system comprising the Project is fully operational and meets the specified performance criteria. The conduct of these Integrated System Tests shall include a period of Trial Running. The Contractor shall be in attendance as a requirement during the Defects Liability Period.



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6.5 Trial Running

Following satisfactory completion of the System Acceptance Tests and the Integrated System Tests the Employer will commence an extended period of Trial Running to prove all technical systems in timetable operation, to allow all technical systems to settle and to train staff in working procedures.

Trial Running shall include:

- The regular circulation of a full complement of trains as required for 24 hours of scheduled service, including peak demand. Intentional disruption of the service shall be included (e.g. extended dwell times, vehicle, power systems, ATP, ATO, point operation and train detection failures) in order to check operational stability and safety of the system and the effectiveness of the supervisory control software employed in reducing the effect of such disruptions;
- The determination of the actual headway achieved at each station for all specified routes and including intermediate reversing movements and movement into and out of the depot.

The Contractor shall allow for attendance over the whole of this period, which may be expected to include maintenance and repair activities and also further opportunity for technical staff training.

6.6 Records of Tests

Records of Tests carried out shall be kept by the Contractor and a report and all Test results shall be submitted to the Engineer no later than 30 days after completion of the Test. In addition to any other requirements, the report shall contain the following details:

- material or part of the Works tested;
- location of the part of the Works;
- place of testing;
- date and time of tests;
- technical personnel supervising or carrying out the tests;
- equipment used and method of testing;
- readings and measurements taken during the tests;
- test results, including any calculations and graphs;
- other details stated in the Contract.

6.7 OPERATION & MAINTENANCE

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